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Machine Learning Model To Predict the Pet Adoption Likelihood

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***Abstract*—Veterinary hospitals and private NGO'S like (PFA, Animal-Aid-Unlimited, Friendlies SECA, Blue cross of India, Wildlife trust of India etc.) are committed to saving and restoring the health of animals, teaching the community about responsible animal care, and championing animal protection., which is a tremendous work towards these animals those requires help and support, and towards the society also. In order to help these organizations and hospitals for the welfare of animals that requires human assistance and which are left over to these NGO's and Veterinary hospitals we have developed a model that will predict the Adoption Speed of these Pet animals by the use of machine learning algorithms and utilizing machine learning techniques which are applied to a pet's dataset provided from Kaggle to extract the prime clinical attributes and predict the adoption speed. We implemented three different machine learning approaches, which are Support Vector Machine (SVM), Logistic- Regression and Adaptive-Boost. Taking into account the numerous categorization methods for machine-learning Adaptive-Boost gives the highest accuracy of 91.54%. This statistical algorithm is helpful for predicting the adoption speed of pets which will be helpful for these animals to find their new home what they deserve**

Keywords—Pet adoption, SVM, Adaptive-Boost, Logistic-Regression, Label Encoder, Machine-Learning, Animal-Welfare.

I. Introduction

Indians are progressively processing a deep bond with their pets which are considered as members of their households. People started to pay greater attention towards the suffering endured by certain animals, inspiring large number of admirers of animals to take up Alley cats or canines as companions in India [1]. But as every coin has two sides the other side also says that due to the change in the thinking of the society and growing urbanization the exchange of pets, buying of new pets became very often and increasing buying of new pets tends old pets to be sell and often been given to various NGO's and veterinary hospitals there could be many reasons for the owner to leave the old pet which may be diseases, and some want to switch to new pet. With this research paper we are going to make many pets animals which are being given disciplinary hospitals be the part of their new family using Machine Learning algorithms that predicts how quickly pet gets adopted, which will relief the burden over these hospitals of managing the pets and finding out easily that which pet has how much time to be sold and meet its new family. The algorithm considers various factors such as animal's type, breed, weather the animal is vaccinated or not, colour and many more related descriptions and attributes which are dependent on the pet's adoption speed. We will explore the traditional Machine Learning Methods (Logistic Regression, SVM and Adaptive Boost) for speedy prediction of adoption of the pet animals that are left in hospitals and NGO" s. In this we will be developing strategies to increase overall adoption rates as well as speed of adoption and its early prediction by creating an accurate model that can tell the relation between key attributes such as (Pet ID, Color, Age, Time-Spend-In- Shelter) which we are using in our dataset. We also have used a **Label Encoder** machine from scikit learn to convert categorical values into numeric values because these machine learning algorithms work with the numeric values as they are enumerated on mathematical formulas. Main objective is to highlight and determine the contributors that influence the adoption speed. This knowledge will be the key to helping these animals to find the home that they deserve. Our objectives of this research are as follows.

(i) To develop a predictive model that accurately estimates the time frame with which the pet is going to be adopted so that it can be helpful for veterinary hospitals and NGO's.
(ii) Improving their adoption rates by knowing the attributes that affect the pet's adoption.

II. Literature Survey

Researchers explored different Machine Learning tools to understand what factors influence pet adoption. They applied these tools to data sets containing pet images and descriptions to predict pets with the highest chance of being embraced. As we explored the different research papers related to our topics, we found that many researchers have used different Machine Learning algorithms and image recognition techniques as mentioned by Azizi, et al [2] in his paper, the author has used DNN's to group kitties and canines with 85.92% precision in training and 98.18% accuracy in testing, and author has also used other facial key points detection methods to raise adoption propensity for the pets to be listed among the top ten suggestions which are going to be adopted.

Zhang, et al. [3] in their work, used five ML algorithms such as Logistic-Regression, Naïve Bayes and SVM, Decision tree, random forest, Gradient-Integral etc. Among all these the Random Forest and Gradient- Integral performed with best accuracies 0.392 and 0.385 respectively where in case of Deep Learning algorithms they used three algorithms namely Fully Connected Neural Network, LSTM, Combined Neural Network out of which Fully Connected Neural Network got the best accuracy 0.396 on the test dataset.

Bradley, et al. [4] contrasted four ML learning prototypes comprising of Logistic-Regression, ANN, random forest and

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Gradient-Boosting. Author acquired the data from various sources on the internet, as well as animal refugee camps across United States. Then the data collected was gone through preprocessing and cleansing and to check the effectiveness three metrics were used using recall, precision and F1 Score and outcomes demonstrated that gradient Boosting executed the best.

Marinho, et al. [5] throughout their investigation utilized two distinct ML methods that are Supervised machine Learning and regression in which Supervised machine Learning consists of working with the data that has already been labelled, predicted or classified as training model and categorize and forecast the new unseen data and results, and Regression is used for working with the continuous values that are passed to the model as input and predict the future outcomes. The author has also used the Deep Learning techniques such as CNN and ANN. CNN is explored during training phase and used by the author for reducing the data in a matrix, thus simplifying models' capability of learning in the dense layers, whereas ANN is used with the highest success rate for the development of an application for pet recognition by passing the data several times through the layers of neurons.

Ye, et al. [6] in this paper proposed and Extreme Learning Gradient Casting (ELGC) framework for target variable prediction. ELGC leverages a multimodal representation constructed from various data sources, including images, natural language, metadata, and custom-derived aggregation features. The author employs dimensionality reduction techniques on these features, followed by training with establishment of machine learning algorithms like Light GBM, XGBoost, and Neural Networks (NN), and then the prediction forms these individual models are then combined as additional feature for a final prediction using Ridge Regression. To optimize model hyperparameters, we utilize a combination of Bayesian optimization and cross-validation.

(Chen, 2023) [1] during the course of their investigation applied various machine learning algorithms to their dataset, training and testing data was divided in the ratio of 8:2. The algorithms used by the author are SVM, Random Forest, BP Neural Network, Light GBM and the results after doing the fivefold cross-validation standout Light GBM algorithm with the highest accuracy of 0.7302 and after the Bayesian optimization the accuracy has increased to 0.8036 and the rest of the algorithms got the lesser accuracy mentioned as SVM with 0.4892, Random Forest with 0.6911, BP Neural Network with 0.7134 accuracy.

Parte's work [7], utilized four Predictive modelling classifiers together with feature engineering and vectorization. K-fold cross-validation was used on the Austin Animal Centre dataset to assess Logistic Regression, Neural Networks, XGBoost, and Random Forest models. XGBoost outperformed the other algorithms, with an accuracy rate of 65.33%.

Powell, et al. [8] in their study found that considerable proportion of dogs and cats returned back to the animal rescue centres after adoption which can be distress for both owner and animal, so they did their analysis and developed model that points out or predicts the features and reasons of return. return they implemented Binary logistic regression technique and models to describe the likelihood and predict the characteristics as a result the found that Behavioural issues were reason with the percentage of 36.1%. In 18.3% of cases, incompatibility with household pets was the main cause of dogs being returned to animal rescue centres. This factor was even more common in cats 22.0% with behavioural issues coming in second with 19.7%.

Y. Wang and Y. Liu, [9] in their research scored the cuteness of the stray animals because most of the customers consider the cuteness of the animal to be the priority to adopt it. And for scoring the cuteness of an animal it is important to have characteristics of that animal and image will make the work simpler to understand the concept of the description that how pet looks which requires a large dataset of animals with their proper description, so the author has leveraged a dataset from the Kaggle competition "PetFinder.my," the researchers created a prediction prototype using a dataset that included both elaborated and Graphic data on the animals. This model considered an animal's traits to assess its level of beauty. The model's output was evaluated using Root Mean Squared Error (RMSE); the test data produced a score of 17.71876. This finding suggests that the model, which used both visual and verbal input, was effective in predicting levels of adorability. The model's performance decreased to an RMSE of 17.76449 when the meta-data was deleted, which highlights how important the meta-data is.

W. K. Cheng, W. C. Leong, [10] in their research developed a clever house setup which is a smart recommendation module which contain a program for forecasting the actions of canines and an interaction setup using mobile devices to establish a connection with canine companions from various homes. They have also performed dog emotions recognition and canine growling and acknowledgement to predict the dog's behaviour additionally Resnet configurations were applied to acknowledge and identify canines' feelings and growling pattern. As a result, the comparison of performances between hyperparameter batches the batch 16 got the accuracy of 73.75% when more data sample size was used and batch 32 got the accuracy of 68.75% used with more data sample size.

H. Kim and N. Moon, [11] in order to overcome the problem of information partiality that was present in earlier studies, Kim and Moon [11] created a categorical to numerical generative adversarial network. To increase information variety, their approach made use of diverse- dimensional time series reasoning, using the information gathered from the wearable device the data was employed from multi-axis-sensor system, accelerator- type sensor, gyroscopic instruments and magnetic sensors. And finally, the model performed exceptionally well with a 97% accuracy rate.

M. Nakamura, N. Dhand, [12] in their study examines the influence of visual features on the acceptance speed of canines posted on Pet Rescue. Australia's biggest internet-based pet adoption was examined by authors. They found substantial associations between certain visual qualities and length of stay in their research of 8332 dog profiles. According to the study, adoption rates were higher for canines with large ears, canines with ebony fur, dogs who have their mouth closed etc. These findings go counter to other studies

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that claimed canines that appeared in settings resembling homes were more appealing. The result suggests that a large number of adopters over the internet could be drawn to animals that are thought to require more care.

In 2013 research, L. H. Ly and A. Protopopova, [13] looked at what makes pet-uniting with families online-like "AdoptaPet.com's" "rehome" service successful. The goal of this strategy is to keep animals out of shelters, where they could feel stressed. More than 148,000 dogs and cats were reunited using the site, whose information the researchers examined. It was discovered that 85.7% of kitties and 87.1% of canines were effectively redirected from shelters. It's captivating to note that the initial owner kept the animal in 37.8% of canines' cases and 35.3% of kitties' cases. The results of logistic regression study showed that less substantial and younger animals had a higher chance of finding new homes.

E. Weiss, M. Slater and L. Lord, [14] conducted a nationwide over the phone survey from September to November 2010, on having pets' ownership and rate of recuperation was the goal of the study. Out of the 6,900 houses they reached out to, 4,029 of them completed the first contact. Of these 2,587 homes that had owns canines and kitties in the previous five years answered the survey, yielding a 64% rate of reaction. According to the poll 75% of lost kitties and 93% of lost canines were finally located and recovered. Searches across the locality and the pet coming home on its own were common recovery techniques.

M. P. Than [15] in their work subsequently created the MI3 (Myocardial-ischemic-injury-index) machine learning model to forecast type 1 myocardial infraction. The prototype was assessed on an additional 7998 people who had a presumed cardiac arrest after being prepared on data from 3,013 patients. The MI3 algorithm uses the patient's age, sex, and cardiac troponin levels to provide a score that ranges from 0 to 100.

Table 1. shows the datasets, machine learning algorithms used, the accuracies and results of various researchers. This above-mentioned table shows the datasets, machine learning algorithms used, the accuracies and results of various researchers.

TABLE I.WORK DONE BY PREVIOUS RESEARCHERS

Ref.	Table Demonstrating previous Research Results		
	Dataset	***ML Classifier / Technique***	***Accuracy and Results***
(Chen, 2023)	American Society for the Prevention of Cruelty to Animals (ASPCA) Dataset.	SVM, Random Forest, BP Neural Network and Light-GBM.	(0.489), (0.6911), (0.7134), (0.8036) respectively.
(Azizi, et al)	COCO BCSPCA, and Standford Dog Dataset	DNN's	(85.92%) accuracy in training and (98.18%) accuracy in testing.
(Marinho, et al)	Dataset of Images Collected.	VGG16 network	93% accuracy.
(Zhang, et al)	.PetFinder and Kaggle	Random Forest, Gradient Boosting and Fully connected Neural Network	(0.392), (0.385) and (0.396)
(Bradley,et al)	Online Sources and Numerous Shelters across the United States	Logistic Regression, Artificial Neural Network, Gradient-Boosting	As the result Gradient Boosting algorithm performed best overall.
(Ye, et al)	PetFinder.my Adoption Prediction	Light GBM, XG Boost and Neural Networks	Ensemble Learning improved the accuracy of prediction
(Parte's work)	Austin Animal center outcome Dataset	XGBoost	65.33%
(M.Nakamura , N.Dhand)	PetRescue Website	Image Analysis	57%
(M. P. Than)	Multiple Centers	MI3(Myocardial ischemic-injury-index)	Identifying patients as low risk (99% sensitivity) and high risk (75% positive)
(L. H. Ly and A. Protopopova)	Adopt a Pet Rehome Website ("Rehome")	Logistic Regression	Overall percentage 87.1% and 37.8% of dogs and 35.3% of cats were retained by their original owners.
(H. Kim and N. Moon)	The Prediction model dataset	Text-to-Numeric Generative Adversarial Network (TN-GAN)	97%
(E.Weiss, M Slater and L Lord)	A professional survey research	Logistic Regression	As a result, 6,996 calls were made
Y. Wang and Y. Liu	Kaggle Dataset	Image Regression Model called PETS-SWINF	RMSE (Root Mean Squared Error) loss of the proposed model on the test dataset is 17.71876 but 17.76449 without metadata

III. ARCHITECTURE

Our architecture model consists of four main stages from which the first one is the data collection phase the data is collected from the Kaggle database site. After that we come to the second stage of data preprocessing where we have done data cleaning and data normalization in which we have removed many outliers and missing values so that data can become suitable for our model. After that comes our third stage un which we have done data exploration in which we have extracted the most important insights and explored our data sets in detail this stage includes data visualization through bar graphs, box plots, relationship graphs and many other statistical data. After the third stage comes our fourth stage which comprises of three algorithms that are mentioned as

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follows (SVM, Logistic Regression and Adaptive-Boost). Figure 1 shows the architecture of our model.

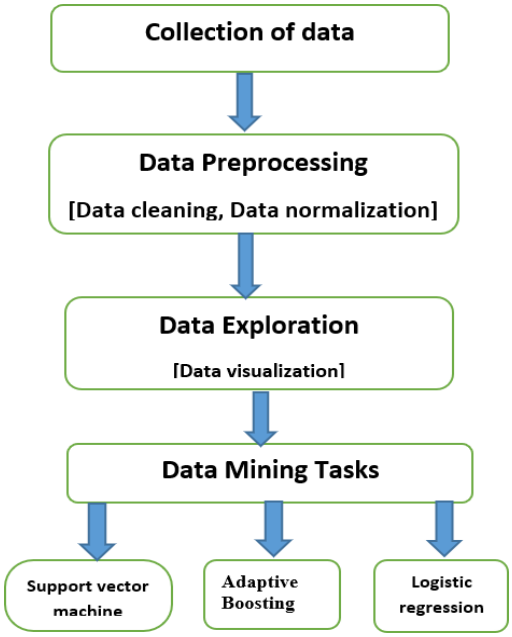


Fig. 1. Architecture

IV. Methods

A. Data Collection

Our research paper used openly accessible Pet-Animals dataset from Kaggle. The Dataset contains attributes like Pet-ID, Pet-Type, Breed, Age, Colour, Size, Weight, Vaccinated, Health-Condition, Time in Shelter, Adoption Speed etc. as done by [16], [17] where the Adoption Speed is considered as dependent variable and our dataset contains approx. 2008 rows with four categories of pets which are Cat, Dog, Bird and Rabbit and these attributes are used for the prediction of the pet adoption speed.

B. Data Preprocessing

Data Preprocessing consist of data cleaning and data normalization which are one as our data contains many missing values and some values that are not feasible such as in one case in our dataset age of a pet is 408 months which is approximately to 34 years and no pet animal such as Dog or Cat can live up to this much age so the Dat Preprocessing was a necessary step to normalize the data and remove the outliers.

C. Feature Choice and Methodology

Choice of features is made to pick up the most informative features as done by [18]. From our dataset Feature Choice helps in decreasing the input variables and expanded the precision of the prototype and its output. Our dataset contains some of the categorical features such as Colour, size, Breed (Description of Pet) these categorical feature needed to be converted in numeric features for the better performance ad higher accuracy rate of our machine learning model so we have used *Label Encoder* class from scikit learn to convert these categorical values into numeric values.

D. Data Exploration

Data Exploration enables us to understand the data characteristics by identifying the hidden patterns, outliers, distributions, and relationship among variables as noted by [19].

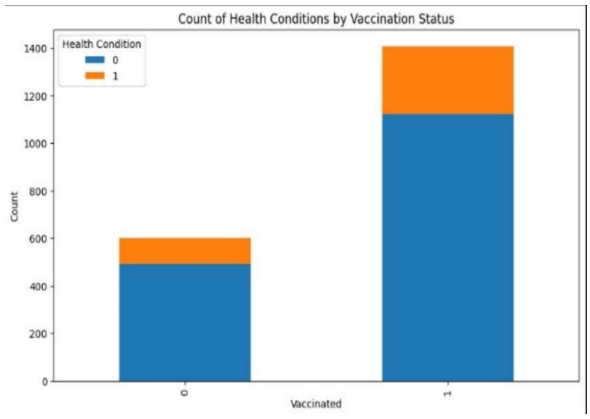


Fig. 2. Showing Correlation between two variables

So, the data-visualization becomes important step as the knowledge gained by data exploration and visualization is important for the model's performance, So this bar graph in Fig. 2. Shows a potential correlation between vaccination status and the presence of health conditions, Vaccinated status is represented in blue color and pets having health conditions or issues are represented by orange color. As we can see in the bar graph that count of the pets which are vaccinated (denoted by 0) is lower as compared to the pets which and not vaccinated (denoted by 1) and there we can also analyze that pets that are vaccinated have lower risk of health issues or conditions as orange portion is generally lower. On the other hand, in the case of not vaccinated category of pets the orange portion is higher that means health problems are more prone to the pets [20] which are not vaccinated.

Swarm Plot Description:

In the Swarm plot the X-axis contains health issues (0) and no health issues (1).

So, from the Swarm Plot we can say that the wider spread of animals having health conditions indicates that they have spent more time in the Shelter as compared to the animals with no health issues which are adopted faster.

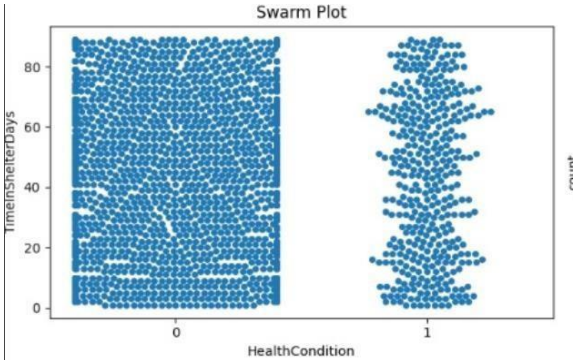


Fig. 3. Showing relation between Healthconditions and time spend in shelter and care centers.

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This can depend on many reasons for example listed below:

- *The Severity of the Health Condition*: More serious conditions might require additional care or treatment, extending the shelter stay[21].
- *Age and Breed Factors*: Younger animals might be adopted faster, while older animals or specific breeds might have longer wait times.
- *Behavioural issues*: Animals with behavioural issues might be more challenging to place, leading to longer shelter days.
- *Adoption Rates*: Fluctuations in adoption rates can influence the overall time animals spend in shelters. High adoption rates might reduce the average stay in shelter overall.

V. MACHINE LEARNING ALGORITHMS

We have utilized three machine learning methods namely SVM, Logistic Regression and Adaptive Boost.

A. SVM

The Best hyper plane or decision border that maximally divides points of data into distinct classes, is found using Support Vector Machines (SVM). The goal of this hyper plane's positioning is too optimum the margin that is, the separation between it and the support vectors, or closest points of the data for each class. Using a Linear Kernel, SVM was used in our study to find the best linear line to split the pet cases into "adopted" and "not-adopted" classifications in our lower-dimensional dataset. In order to build the final classification model, the computer identified the pet instances in both categories that were closest to this decision line.

B. Logistic Regression

In Logistic Regression we have divide the data into the dependent variables and independent variables (x) and the mathematical formula takes the linear combination of the independent variables weighted by the respective coefficients (β), the Linear combination is then passed through the logistic function (1 / (1 + e^(-x))) which squashes the value between 0 and 1, the resulting value, h(x), represents the predicted probability of belonging to the target class (often coded as 1). Here is the formula used in Logistic Regression.

$$P = 1 / (1 + e^{(-z)}) \quad (1)$$

Where:

- P is the probability of a pet being adopted ($0 <= p <= 1$).
- e is the base of the natural logarithm (approximately 2.71828).
- z is the linear combination of predictor variable and their corresponding coefficients.

$$z = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

- β_0 is the intercept.
- $\beta_1, \beta_2, \ldots, \beta_p$ are the coefficients for the predictor variables $X_1, X_2, \ldots, X_p$.
- The equation calculates the probability of adoption (p) based on the values of the predictor variables ($X_1, X_2, \ldots, X_p$) and their associated coefficients ($\beta_1, \beta_2, \ldots, \beta_p$).
- The linear combination of predictors (z) is transformed into a probability value between 0 and 1 using the logistic function 1 / (1 + e^(-z)).

C. Adaptive Boost

The reasons for using Adaptive Boost is our machine learning adoption predicting model is that as from the references we get to know that in the related work the Adaptive Boost machine learning algorithms is performing with the best accuracy and results. The other reason for using Adaptive Boost is that it is a classifier and a boosting algorithm that changes weak beginners to powerful learners.

$$H(x) = \text{sign}\ \Sigma_{t=0}\ \propto\ h_t\ (x)\ ) \quad (2)$$

In the above Equation:

- H(x) signifies the quantity of input instruction data.
- h_t (x) connects to the feeble identifier t output where x is input.
- α_t represents quantity shared with the identifiers.

n excellent style manual for science writers is [7].

VI. TRAINING AND EVALUATION

Training and evaluation is carried out as shown in table 2. Mentioned below to complete the accuracy on data by splitting it into two parts that are training splits and testing splits. Additionally, cross validation is done to gauge a model's performance on unseen data.

TABLE II. RESULTS OF TRAINING AND TESTING

Machine Learning Classifier	Parameters	Parameters Used
SVM	Kernel Solver	Linear libSVM (library)
Logistic Regression	Evaluation Data Splitting One-Hot-Encoding	Accuracy score test_size (default = 0.2) sparse (default = false)
AdaBoost Classifier	No of estimators	50

VII. RESULTS AND DISCUSSIONS

A. SVM

When the SVM model was tested using the test dataset, its accuracy percentage was 88%. A 10-fold cross validation approach was utilized to evaluate the model's capacity to make generalizations [22] to previously encountered data. To do this, the dataset was divided into 10 equal sized sections, nine subgroups were utilized to instruct the prototype in each iteration, while remaining subgroups were used for assessment. Ten iterations of this method were carried out, using a new subset as the test set each time.

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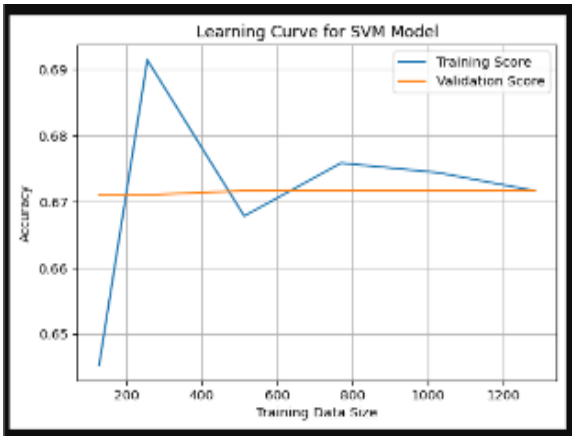


Fig. 4. Showing the Learning Curve for SVM Model

B. *Logistic Regression*

Our classification model employed a multinomial logistic regression approach to handle the response variable having five categories influenced by the work of [23]. This model estimates the probability distribution of the response's variable belonging to each class, considering the input features and the learned model parameters [24]. As by using the Logistic Regression we achieved accuracy percentage of 87 %.

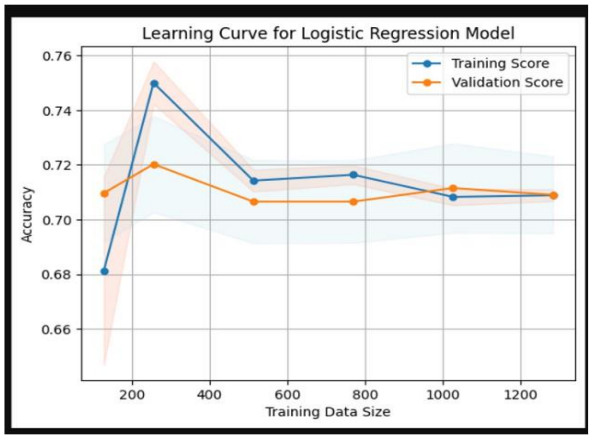


Fig. 5. Showing the Leaning Curve for Logistic Regression.

C. *Adaptive Boost*

Where other such as Logistic Regression, being powerful but can struggle with the complex relationships between features and the target variable (adoption speed) [27]. Adaptive Boost offers a mathematical advantage on this context by creating an ensemble model that iteratively refines predictions and giving us the highest accuracy score of 91%. As from the study of the various researchers such as [25], [26] Adaptive Boost performs better than the other algorithms because Adaptive-Boost assign weights to each data point in our dataset. Initially, all points have equal weight [28]. In each iteration, a weaker learner is built, mathematically, the error rate of this weak learner is calculated and the weight of data points misclassified by the weak learner are increased. This ensures the next learner focuses on these harder cases. This iterative process leads to a series of weak learners that, when combined, become a string ensemble model, mathematically the contribution of each weak learner is determined by its weight [29] and calculated based on its error rate.

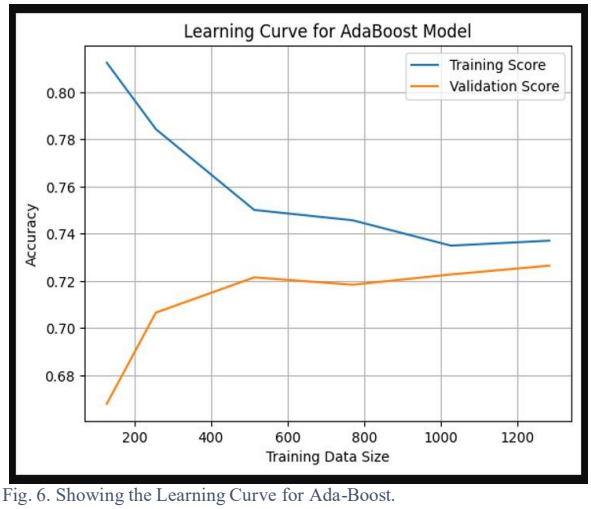


Fig. 6. Showing the Learning Curve for Ada-Boost.

VIII. Conclusions

Our evaluation on all the 3 machine learning algorithms on the basis of accuracy and F1 Score we suggests that the Adaptive-Boost-Classifier [30] emerged as the strongest performer among its competitors which are SVM, Logistic Regression. It achieved the highest values, demonstrating superior ability to distinguish between different adoption outcomes compared to the other two models.

IX. Future Work

Text heads As this field mostly unexplored there is huge aspects of future work can be done to improve this model's precision by collecting and utilizing the longitudinal making it more precise to predictions. This model can be combined to another model to create a hybrid and robust model. Can be done changes with what additional factors beyond adoption speed could be incorporated into the prediction such as Hospital management can use the system to forecast staffing needs based on predicted adoption rates

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